

DP Computer Science: A 3.4.2 Data Warehouses (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the **primary objectives** of data warehouses in data management and business intelligence
 - Describe the role of **append-only** data in ensuring efficient data storage and analysis
 - Describe the role of **subject-oriented** data in ensuring efficient data storage and analysis
 - Describe the role of **integrated** data in ensuring efficient data storage and analysis
 - Describe the role of **time-variant** data in ensuring efficient data storage and analysis
 - Describe the role of **non-volatile** data in ensuring efficient data storage and analysis
 - Describe the role of data **optimised for query performance** in ensuring efficient data storage and analysis
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Key Vocabulary

Term	Definition
Data warehouse	A specialised database designed for storing and analysing large volumes of historical data to support business intelligence and decision-making
OLTP	Online Transaction Processing — systems that handle day-to-day transactions (e.g., order processing, banking)
OLAP	Online Analytical Processing — systems designed for complex analysis and reporting (e.g., business intelligence, trend analysis)
Append-only	Data is added to the warehouse but never modified or deleted; changes are recorded as new records

Term	Definition
Subject-oriented	Data is organised around key business subjects (customers, products, sales) rather than application functions
Integrated	Data from multiple sources is combined into a consistent format
Time-variant	Data includes a time dimension, allowing historical analysis and trend identification
Non-volatile	Data is never deleted or modified once entered; it is read-only for analysis
Query performance	Data warehouse structures are optimised for fast query execution (e.g., indexes, summarised tables)
ETL	Extract, Transform, Load — the process of moving data from operational systems to a data warehouse

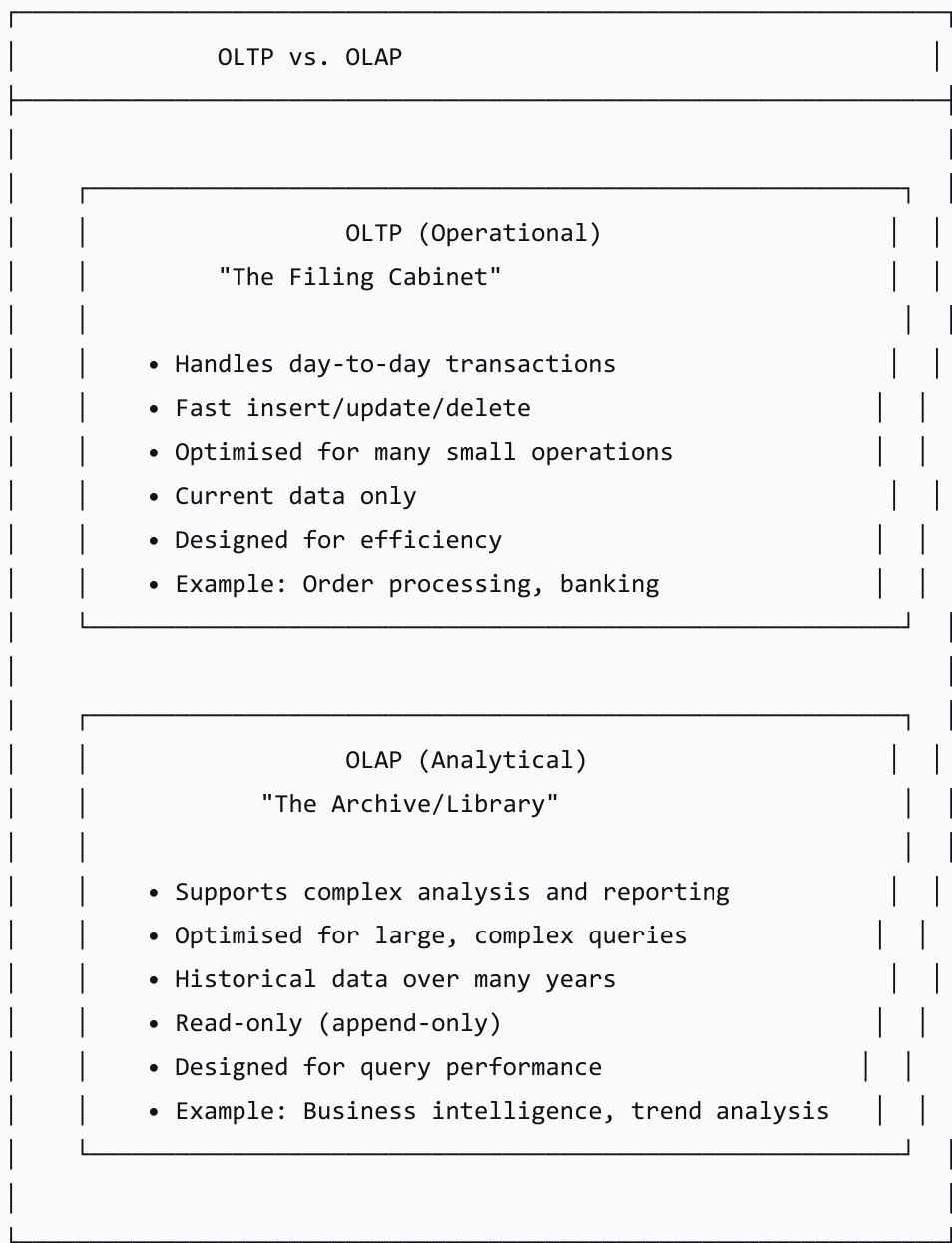
Relevant ATL Skills

Skill	Description
Thinking	Critical thinking (analysing purpose and benefits); understanding (interpreting concepts); applying (considering how characteristics facilitate storage and analysis)
Communication	Explaining concepts; participating in discussions
Research	Comprehending definitions and roles

2. Introduction: Why Do We Need Data Warehouses?

Operational vs. Analytical Systems

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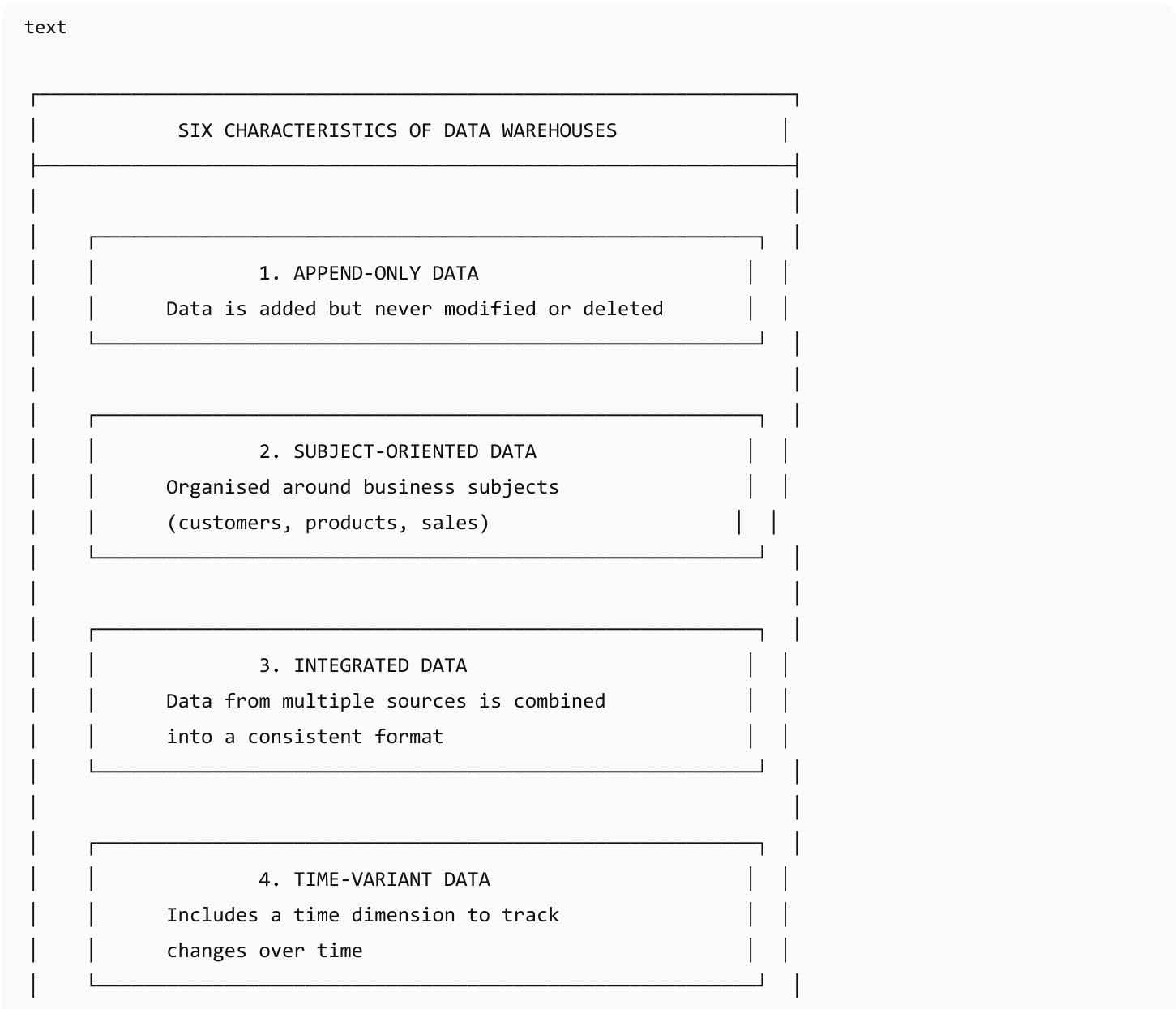
Why Not Just Use the Operational Database?

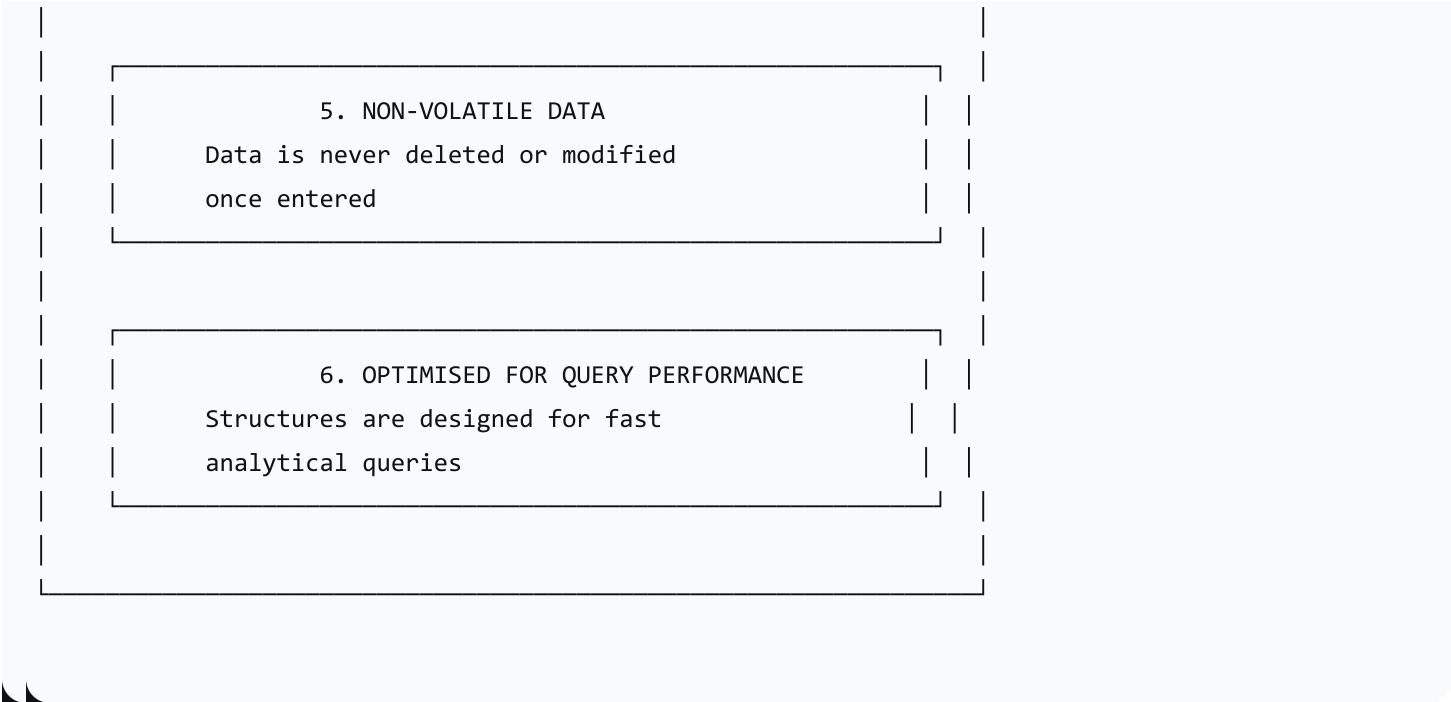
Reason	Explanation
Performance	Complex analytical queries would slow down daily operations
Data Structure	Operational databases are optimised for transactions, not analysis

Reason	Explanation
Historical Data	Operational databases typically store only recent data
Integration	Data warehouses combine data from multiple sources
Read-Only Access	Warehouses provide a stable, read-only view for analysis

3. The Six Key Characteristics of Data Warehouses

Overview





4. Detailed Explanation of Each Characteristic

1. Append-Only Data

In a data warehouse, data is never modified or deleted. Changes are recorded as new records, preserving the complete history.

Aspect	Description
Definition	Data is only added (appended) to the warehouse; no updates or deletes
Why it Matters	Preserves historical context; allows accurate trend analysis
How it Works	Instead of updating a record, a new record is added with a timestamp
Example	Customer changes address → Add a new record with the new address and date
Advantage	Explanation
Complete History	Every change is preserved

Advantage	Explanation
Auditability	Can track exactly when changes occurred
Trend Analysis	Can analyse patterns over time
No Data Loss	No risk of accidentally deleting historical data

Analogy: A filing cabinet where you never throw anything away—you just add new papers to the folder.

2. Subject-Oriented Data

Data in a warehouse is organised around key business subjects rather than application functions or daily transactions.

Aspect	Description
Definition	Data is categorised by business subjects (customers, products, sales)
Why it Matters	Makes data easier to query for business analysis
How it Works	Data is structured around the things that matter to the business
Example	Sales data is organised by customer, product, and sales period

Subject	Description	Examples
Customer	Information about customers	Customer ID, name, address, segment
Product	Information about products	Product ID, name, category, price
Sales	Information about transactions	Sale ID, date, amount, customer, product
Location	Geographic information	Region, country, city

Analogy: A library organised by topic (history, science, fiction) rather than by who donated each book.

3. Integrated Data

Data from multiple sources (operational systems, external data) is combined into a consistent format in the data warehouse.

Aspect	Description
Definition	Data is collected from multiple sources and standardised
Why it Matters	Enables analysis across different systems and data sources
How it Works	ETL (Extract, Transform, Load) processes convert data into a common format
Example	Sales data from online and in-store systems is combined

Integration Challenges:

Challenge	Solution
Different formats	Standardise data into a common format
Different currencies	Convert all currencies to a common base
Different IDs	Map different IDs to a consistent identifier
Inconsistent naming	Standardise naming conventions
Missing data	Handle missing values appropriately

Analogy: Translating multiple languages into one common language so everyone can understand.

4. Time-Variant Data

Data in a warehouse always includes a time dimension, allowing historical analysis and trend identification.

Aspect	Description
Definition	Every piece of data has a time component (date, timestamp)

Aspect	Description
Why it Matters	Enables analysis of changes over time
How it Works	Timestamps are added to all data records
Example	Sales data includes the date of the transaction

Time Dimensions:

Dimension	Description	Example
Date	The day of the event	2026-01-15
Time	The time of day	14:30:00
Year	The year of the event	2026
Quarter	The quarter of the year	Q1, Q2, Q3, Q4
Month	The month of the year	January
Week	The week number	Week 3

Analogy: A photo album organised by date—you can see how things changed over time.

5. Non-Volatile Data

Data in a warehouse is never deleted or modified once entered. It is read-only for analysis purposes.

Aspect	Description
Definition	Data is permanent and never changed
Why it Matters	Preserves historical context; ensures accurate analysis
How it Works	No DELETE or UPDATE operations; only INSERT
Example	Customer data is never deleted, even if they leave

Advantage	Explanation
Historical Accuracy	Past data remains unchanged and reliable
Auditability	Complete audit trail of all data
Predictive Analysis	Can analyse long-term trends
Regulatory Compliance	Meets data retention requirements

Analogy: A public archive where documents are stored permanently and never removed.

6. Optimised for Query Performance

| Data warehouse structures are designed for fast execution of complex analytical queries.

Aspect	Description
Definition	The warehouse is structured for fast data retrieval
Why it Matters	Business intelligence queries often process millions of records
How it Works	Use of indexes, summary tables, and denormalisation
Example	Pre-calculated monthly sales summaries

Optimisation Techniques:

Technique	Description
Indexes	Speed up searching and filtering
Summary Tables	Pre-calculated aggregations (daily, monthly totals)
Star Schemas	Fact and dimension tables for efficient joins
Column Storage	Data is stored by column, not by row (faster for aggregations)
Partitioning	Data is split by date or other criteria

Analogy: A restaurant menu organised into sections (appetisers, mains, desserts) with pictures—it's easier to find what you want.

5. Summary of the Six Characteristics

Characteristic	Key Idea	Benefit
Append-Only	Add new records; never modify or delete	Preserves complete history
Subject-Oriented	Organised around business subjects	Easier analysis and reporting
Integrated	Combined from multiple sources	Holistic view of business
Time-Variant	Includes time dimension	Trend and historical analysis
Non-Volatile	No updates or deletions	Stable, accurate data
Optimised for Performance	Designed for fast queries	Fast response to complex queries

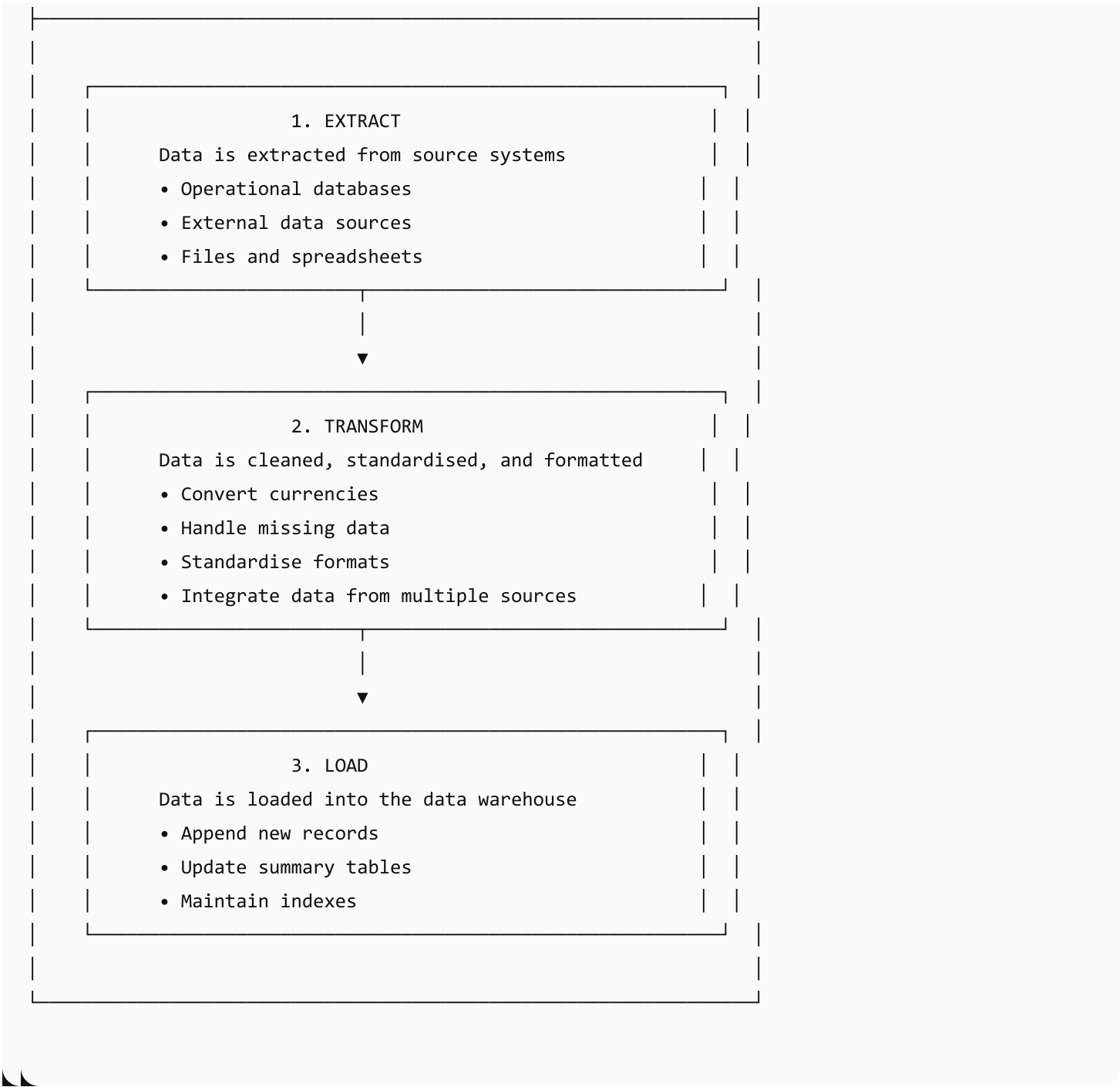
6. The ETL Process

What is ETL?

ETL (Extract, Transform, Load) is the process of moving data from operational systems to a data warehouse.

ETL Steps





7. Data Warehouse vs. Operational Database

Feature	Operational Database (OLTP)	Data Warehouse (OLAP)
Purpose	Day-to-day transactions	Analysis and decision-making
Data	Current data	Historical data

Feature	Operational Database (OLTP)	Data Warehouse (OLAP)
Operations	INSERT, UPDATE, DELETE	Append-only (no updates/deletes)
Queries	Simple, specific	Complex, aggregated
Users	Operational staff	Management, analysts
Performance	Fast individual transactions	Fast complex queries
Storage	Normalised (reduces redundancy)	Denormalised (improves query speed)
Size	Smaller (GBs)	Larger (TBs)
Example	Order processing system	Sales trend analysis

8. Practical Activity: Building a Data Warehouse in SQLite

Activity Overview

Students will build a simple data warehouse simulation in SQLite, practising each of the six characteristics.

Task 1: Append-Only Data

Scenario: Customer address changes must be tracked historically.

```
sql

-- Instead of updating, add a new record
CREATE TABLE CustomerHistory (
    CustomerID INTEGER,
    Address TEXT,
    EffectiveDate DATE,
    PRIMARY KEY (CustomerID, EffectiveDate)
);

-- Add a new address record when customer moves
```

```
INSERT INTO CustomerHistory VALUES (1, '123 Main St', '2024-01-01');
INSERT INTO CustomerHistory VALUES (1, '456 Oak Ave', '2025-06-15');
```

Task 2: Subject-Oriented Data

Scenario: Create a customer subject view.

```
sql

CREATE VIEW CustomerSubject AS
SELECT
    c.CustomerID,
    c.Name,
    c.Segment,
    COUNT(o.OrderID) AS TotalOrders,
    SUM(o.Total) AS LifetimeValue
FROM Customers c
LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
GROUP BY c.CustomerID;
```

Task 3: Integrated Data

Scenario: Combine sales data from multiple sources.

```
sql

CREATE VIEW IntegratedSales AS
SELECT OrderID, Date, Amount, 'Online' AS Source
FROM OnlineSales
UNION
SELECT OrderID, Date, Amount, 'InStore' AS Source
FROM StoreSales;
```

Task 4: Time-Variant Data

Scenario: Analyse monthly sales trends.

```
sql
```

```
SELECT
    strftime('%Y-%m', Date) AS SalesMonth,
    SUM(Amount) AS TotalSales
FROM Sales
GROUP BY SalesMonth
ORDER BY SalesMonth;
```

Task 5: Non-Volatile Data

Scenario: A customer moves—do not update; add a new record.

```
sql

-- ❌ Wrong: UPDATE Customer SET Address = 'New Address' WHERE ID = 1;

-- ✅ Correct: INSERT INTO CustomerHistory VALUES (1, 'Old Address', '2020-01-01');
--             INSERT INTO CustomerHistory VALUES (1, 'New Address', '2025-06-15');
```

Task 6: Optimised for Query Performance

Scenario: Create a summary table for fast reporting.

```
sql

CREATE TABLE MonthlySalesSummary AS
SELECT
    strftime('%Y-%m', OrderDate) AS SalesMonth,
    COUNT(*) AS OrderCount,
    SUM(Total) AS TotalSales
FROM Orders
GROUP BY SalesMonth;
```

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	8 min	Introduce OLTP vs. OLAP; primary objectives; why we need a data warehouse
Tool Setup	2 min	Open DB Browser for SQLite; orientation on "Execute SQL" tab
Append-Only & Subject-Oriented	8 min	Explain both concepts; students complete Tasks 1 and 2
Integrated & Time-Variant	8 min	Explain both concepts; students complete Tasks 3 and 4
Non-Volatile & Optimised	8 min	Explain both concepts; students complete Tasks 5 and 6
Deep Dive Discussion	11 min	Link tasks back to objectives; reflection questions
Conclusion	5 min	Summary; Q&A

Reflection Questions

Task	Reflection Question
Task 1	Why is appending safer than updating for historical data?
Task 2	How does organising data by subject help business analysis?
Task 3	What would happen if currencies weren't integrated?
Task 4	Why is time-variant data important for trend analysis?
Task 5	How does Task 5 demonstrate non-volatility?
Task 6	Why is the summary table faster for a CEO's dashboard?

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Practical Activity	Evaluate completion and understanding of tasks
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between an operational database and a data warehouse?
 2. What does "append-only" mean in the context of a data warehouse?
 3. Why is data in a data warehouse "non-volatile"?
 4. Give one reason why data warehouses are optimised for query performance.
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11. Differentiation

Level	Strategy
Support	Provide more scaffolding for SQL tasks; break down syntax into smaller steps
Challenge	Explore Star Schemas; consider how ETL pipelines work in the real world

12. Resources

- **Presentation** (Slide Deck)
 - **eBook** (A3.4.2 with examples)
 - **Activity Worksheet** (Tasks for each characteristic)
 - **Activity Worksheet Answers** (Exemplar responses)
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13. Summary: Key Takeaways

Characteristic	Key Point
Append-Only	Never modify/delete—only add new records
Subject-Oriented	Organised around business subjects (customers, products, sales)
Integrated	Combined from multiple sources into a consistent format
Time-Variant	Includes a time dimension for historical analysis
Non-Volatile	Data is permanent; no updates or deletions
Optimised for Performance	Designed for fast complex analytical queries

text

KEY REMINDERS

Data warehouses store HISTORICAL data

Operational databases store CURRENT data

Data warehouses support Business Intelligence

ETL = Extract, Transform, Load

Data warehouses = Append-Only, Non-Volatile

Data warehouses are optimised for complex queries

"A data warehouse is a specialised database designed for analysis—it stores historical data, organises it by business subjects, and optimises it for fast query performance."